

Xiwei Zhao

<https://xiweizhao118.github.io/>

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EDUCATION

- **Aalto University** Helsinki, Finland
Master of Science in Robotics and machine learning; GPA: 3.94 (4.79/5.00) Aug. 2023 – Present
- **Harbin Institute of Technology** Harbin, China
Bachelor of Engineering in Automation; GPA: 3.61 (90.2/100.0) Aug. 2019 – July. 2023

PUBLICATION

- **Video-to-BT: Generating Reactive Behavior Trees from Human Demonstration Videos**
Xiwei Zhao, Yiwei Wang, Yansong Wu, Fan Wu, et al. Submitted to ICRA 2026. [arXiv] [Website]

EXPERIENCE

- **MIRMI, Technical University of Munich** Munich, Germany
Master thesis worker July. 2025 - Present
 - **VLM-Driven Reactive Behavior Tree Generation:** Architected a VLM-based system to translate demonstration videos into structured behavior trees for high-level task monitoring.
 - **Force-Position Hybrid Control for Assembly Tasks:** Implemented adaptive impedance control algorithms enabling < 0.5mm positional error in plug-in operations through dynamic torque/position coupling.
- **Seed Robotics, ByteDance** Beijing, China
Embodied AI Research Intern Jan. 2025 - June. 2025
 - **Enhanced VLM Decision-Making:** Fine-tuned Qwen-2.5VL-7B to improve VLM reasoning in novel environments, boosting task success rate from 40% to 70% and achieving performance on par with GPT-4o.
 - **Multi-Agent Robotic Planning Framework:** Developed a collaborative multi-agent framework based on LangChain for automated robotic task planning, achieving: 90%+ accuracy in high-level assembly procedure generation aligned with manual specifications; 80% reliability in producing fine-grained manipulation commands
- **Intelligent Robotics Group, Aalto University** Helsinki, Finland
Research assistant June. 2024 - Dec. 2024
 - **VLA Models Utilization in Deformable Items Grasping:** Fine-tuned Octo and OpenVLA models for deformable object grasping tasks on IsaacSim. Built a high-precision simulation environment, adapting the models to handle fabrics and plastic bags.
 - **Automatic Dataset Collection in IsaacSim:** Created a plastic bag manipulation framework with Isaac Cortex for automated dataset collection. Implemented behavior tree technique for adaptive subtask transitions, adjusting strategies based on object states.
- **Wireless Lab, Institut national de la recherche scientifique (INRS)** Montreal, Canada
Machine Learning Research Intern, Mitacs-CSC scholarship July. 2022 - Dec. 2022
 - **Machine Learning for Channel Estimation:** Optimized channel models in communication systems and applied machine learning algorithms, specifically the Expectation-Maximization (EM) algorithm, to estimate channel transmission parameters.
 - **Mathematical Modeling:** Developed and refined mathematical models to enhance the accuracy and efficiency of channel estimation in communication systems.

PROJECTS

- **QuantSoftware Toolkit:** https://github.com/xiweizhao118/TS_factor_analyst
Open source python library for financial data analysis and machine learning for finance. Offered an integrated analysis frame with factor calculation and evaluation. Codes are available on GitHub.
- **GPT-based Franka Robot:** https://github.com/xiweizhao118/LLM_Robotics
Based on the GPT-3.5 Turbo model fine-tuning, enabled Franka to comprehend natural language instructions from users. Conducted simulation using ROS and provided complete code on GitHub repository.

PROGRAMMING SKILLS

- **Languages:** Python, C++, JavaScript, Matlab
- **Tools:** VS Code, Git, Latex